

AceFX-AI Research Report

Phase 1-3 Findings and Current State

A plain-language summary of what the project tested, what failed, what remains useful, and what the project is today.



Executive Summary

AceFX-AI is still a research project, not a deployable trading system. No active trading candidate exists today, which means none of the branches tested so far has produced a strategy strong enough to justify promotion into a more serious trading track.

Some internal notes use shorthand such as "Candidate 7 was not created." For an outside reader, the simpler interpretation is: no branch has yet passed the project's promotion standards strongly enough to open the next candidate stage.

- Phase 2 technical-indicator and LightGBM research did not produce a robust tradable edge.
- Phase 3 event research produced one useful EURUSD unemployment-rate volatility feature, but not a strategy.
- Project X is paused.
- RL, demo execution, and live execution remain blocked.

The strongest asset created so far is therefore the research process itself: external-first validation, anti-overfitting controls, pre-registered hypotheses, and explicit PASS / REVIEW / FAIL gates.

Item	Current result
Active trading candidate	None
Tradable strategy ready for use	None
Has any branch earned promotion into the next candidate stage?	No
Phase 2 technical / LightGBM path	Exhausted
Phase 3 event branch	Closed / checkpointed
EURUSD unemployment-rate result	REVIEW; feature-only
XAUUSD PCE result	FAIL
Project X	Paused
RL / demo / live execution	Blocked

How To Read This Report

This document is written for a first-time reader, so a few internal project terms need translation.

- Candidate: a research branch that looks strong enough to justify deeper validation as a possible future strategy. It is not the same thing as a live system.
- Shadow logging: recording what happened around a setup or feature without placing trades.
- PASS: strong enough to justify the next validation step.
- REVIEW: interesting or useful, but not robust enough for promotion.
- FAIL: did not survive the project's validation standards.

The central point is simple: AceFX-AI has built a serious research framework, but it has not yet found a trading strategy strong enough to advance.

Research Philosophy

AceFX-AI now follows a conservative, evidence-driven posture. The goal is to avoid confusing an interesting backtest pattern with a real portable edge.

Principle	Practical meaning
External-first validation	A result that works only on one broker feed is not trusted.
Pre-registered hypotheses	Branches start with explicit rules, not after-the-fact reframing.
PASS / REVIEW / FAIL gates	Every branch must earn the next step.
Anti-overfitting controls	Spread stress, concentration checks, and strict closeout rules are mandatory.
Future-only validation	No strategy can be promoted without forward-looking evidence.

Phase 1-2 Technical Research Summary

Phase 1 and 2 asked a direct question: can standard technical indicators plus tree-based models such as LightGBM produce a stable edge in XAUUSD or major FX pairs?

After progressively stricter validation, the answer was no. Candidate 1 failed its shadow-promotion path. Candidate 4, which looked stronger on the FusionMarkets feed, failed external validation on Dukascopy data. That was a major lesson because it showed the effect was likely feed-sensitive rather than portable.

The project then tested CatBoost, XGBoost, and learned process meta-gates. These branches did not fix the same underlying weakness: results were still not robust enough across newer windows and stricter validation.

- XAUUSD H1 was tested as a lower-cost external-first baseline.
- EURUSD M15 and H1 were tested as major-FX alternatives.
- Broader FX triage scanned more pair / timeframe combinations.

XAUUSD H1 was somewhat less cost-sensitive than M15, but still too weak. EURUSD M15 and H1 were the least-bad branches, yet still failed stability and spread-stress requirements. Broad FX triage also failed to find a pair / timeframe that passed the gate.

Phase 1-2 branch	Final read
XAUUSD M15 initial path	Failed portability and cost checks
Candidate 1 shadow track	Failed later observation windows
Candidate 4 rollover process	Failed external validation
CatBoost / XGBoost variants	Did not fix the same stability weakness
Learned meta-gate variants	Did not beat the simpler Candidate 4 process

Phase 1-2 branch	Final read
XAUUSD H1	REVIEW, but still too weak
EURUSD M15 / H1	Less bad, but still failed stability and spread stress
Broad FX triage	No pair / timeframe passed the gate

Phase 3 News/Event Research Summary

Phase 3 changed the question. Instead of asking whether generic technical structure alone could surface an edge, the project asked whether specific macro events could create clearer, more explainable market behavior.

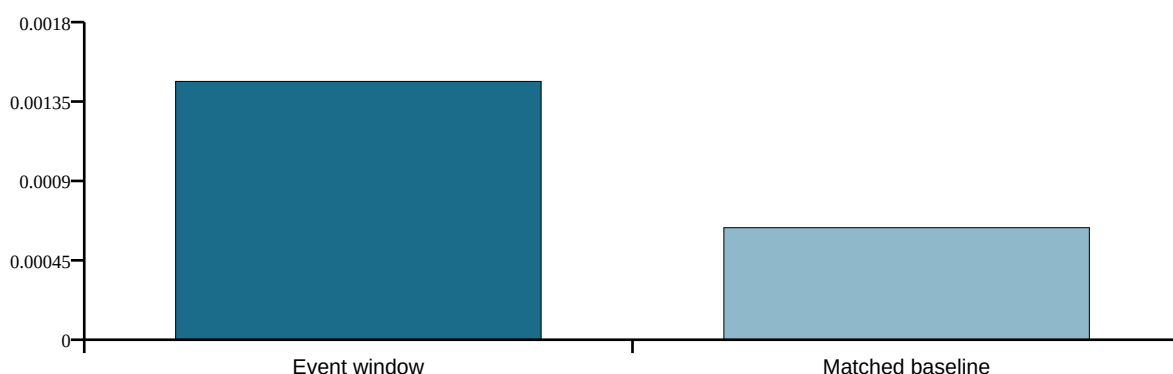
- An MT5 economic calendar exporter and normalized importer were built.
- Multi-year calendar coverage across 2021-2026 was assembled.
- Event alignment diagnostics and event-behavior audit tooling were added.
- Locked hypothesis-test workflows were used to limit post hoc interpretation.

The strongest branch was EURUSD unemployment-rate events. The locked hypothesis stayed at REVIEW rather than PASS, but it did show consistent evidence of elevated post-event volatility in the primary 0 to +1h window. The project therefore froze it as an event-regime / no-trade / volatility feature rather than promoting it into a strategy.

The XAUUSD event branch was mixed. Gap classification remained REVIEW. XAUUSD PCE eligibility passed as a data-readiness gate, meaning there were enough complete windows to justify a locked test. But the locked XAUUSD PCE hypothesis itself failed because the primary behavior was not materially above the matched non-event baseline.

Phase 3 component	Outcome
MT5 calendar exporter / importer	Built
Multi-year event coverage	2021-2026 available
Event alignment diagnostics	Built and used
Event-behavior audit	REVIEW
EURUSD unemployment-rate locked hypothesis	REVIEW
EURUSD unemployment-rate frozen feature	REVIEW; feature-only
XAUUSD gap classification	REVIEW
XAUUSD PCE eligibility for testing	PASS as a data-readiness gate
XAUUSD PCE locked hypothesis	FAIL
Future-only shadow logger	Built for observation only

EURUSD unemployment-rate window: average absolute return vs matched baseline



Most Significant Findings

The most important outcomes are not marketing points. They are the places where the project learned what does not survive stronger validation, plus one feature-level result that may still be useful operationally.

1. Candidate 4 failed external portability.

On the original broker context, Candidate 4 looked like a plausible execution-aware edge. Under external validation it did not survive. This matters because it shows why broker-independent checking is essential.

2. Broad FX triage did not find a winner.

EURUSD M15 and H1 were the least-bad branches, but they still failed once fold concentration and spread stress were enforced. Expanding the search across pairs and timeframes did not uncover a robust edge.

3. EURUSD unemployment-rate events showed elevated volatility.

EURUSD unemployment metric	Value
Event count	64
Avg absolute return, 0 to +1h	0.001464
Matched non-event baseline	0.000635
Volatility expansion vs baseline	3.948x
Leave-one-year-out primary persistence	100%
Timestamp stress	REVIEW

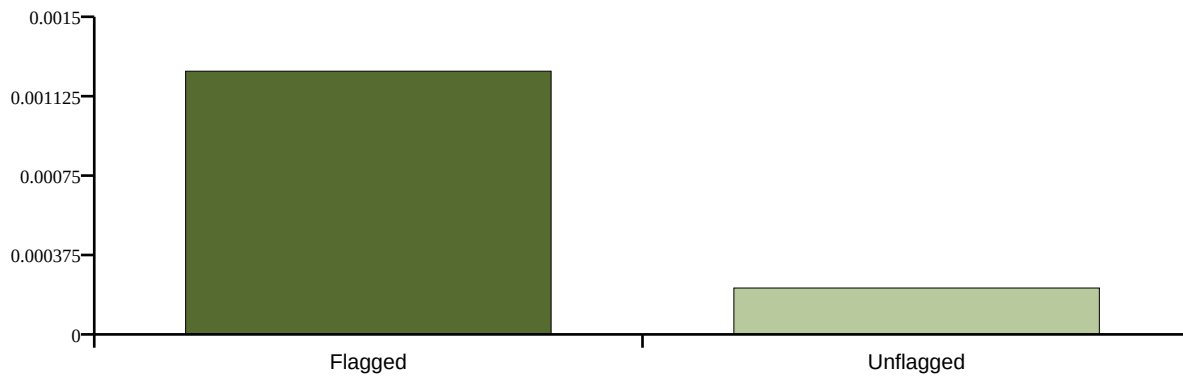
This remains a REVIEW result, not a strategy approval. The main caveat is timestamp sensitivity, and the event group is narrow rather than broad.

4. The frozen event feature is small but historically meaningful.

Frozen feature metric	Value
Flagged candles	256
Total EURUSD M15 candles in audit	191,424
Flagged share	0.1337%
Flagged avg absolute return	0.001243
Unflagged avg absolute return	0.000219
Flagged avg range / volatility	0.002318
Unflagged avg range / volatility	0.000433
Flagged spike frequency	38.67%
Unflagged spike frequency	4.25%

In simpler terms, the feature marks a very small subset of EURUSD candles that were historically more volatile than normal, but the project does not treat that as a trading rule.

Frozen EURUSD feature: avg absolute return of flagged vs unflagged candles



5. XAUUSD PCE failed a locked test.

Eligibility passed, so the idea was not rejected on data-completeness grounds alone. But the locked hypothesis failed because the primary 0 to +1h behavior was not materially above the matched non-event baseline. The project kept the negative result instead of rewriting the hypothesis afterward.

Finding	Why it matters
Candidate 4 failed external portability	A promising broker-specific result did not survive independent validation.
Broad FX triage found no winner	Expanding across pairs and timeframes did not uncover a robust edge.
EURUSD unemployment events showed elevated volatility	This is the strongest feature-level result so far, but it is not a strategy.
Frozen EURUSD event feature is small and selective	It marks only 0.1337% of candles, which makes it more useful as a filter.
XAUUSD PCE failed a locked test	The project kept the negative result instead of rewriting the hypothesis.

Current State

The project remains in a deliberately constrained state. No active candidate exists, no tradable strategy exists, and the only forward-looking operational step is feature-only shadow observation.

Current status item	State
Active candidate	None
Tradable strategy	None
Has any branch earned promotion into a new candidate stage?	No
EURUSD unemployment_rate	REVIEW; feature-only
XAUUSD PCE	FAIL
Project X	Paused
RL / demo / live	Blocked
Next real action	Wait for the next USD unemployment event, then run the shadow logger

What This Project Is Now

AceFX-AI is best understood today as a disciplined quant research framework, a validation engine for trading hypotheses, a research journal and audit trail, and possibly a future foundation for research tooling. It is not a profitable trading bot yet.

What the project is now	What it is not
Disciplined quant research framework	Profitable trading bot
Validation engine for trading hypotheses	Production execution system
Research journal and audit trail	Proof of deployable trading edge
Potential future research SaaS foundation	Marketing-ready AI trading product

Next Steps

- Keep the repository private.
- Continue future-only shadow logging for the frozen EURUSD unemployment-rate feature.
- Cross-check economic calendar timestamps with an independent source.
- Avoid reopening event-group shopping from the existing Phase 3 branch.
- Reopen strategy research only if there is a materially different, pre-registered hypothesis with explicit robustness rules.

Final Assessment

AceFX-AI has generated useful research infrastructure and several important falsification results. It has shown that multiple plausible-looking technical and event-based ideas do not survive stricter validation standards. It has also identified one credible feature-level lead in EURUSD unemployment-rate event windows, but that result remains a feature-only finding, not a tradable edge.

That is an honest and useful place to be. The project is not yet a trading product. It is a disciplined research system that has learned where not to overclaim.